dv33: The 33rd Heaven
H = log N is the Hilbert-Pólya Operator

Version 33 — February 25, 2026 — 33 steps. 33 t■ng tr■i. The complete synthesis.

Abstract. Step 33 is the synthesis. We show that H = log N, the arithmetic Hamiltonian identified in dv24, is the arithmetic quantization of the Berry-Keating operator H_{BK} = xp (1999), which was proposed as the Hilbert-Pólya operator whose eigenvalues are the imaginary parts of Riemann zeros. The classical trajectories of H_{BK} = xp are hyperbolas x·p = const in phase space, which is precisely the hyperbolic plane H. The geodesic flow on H is generated by H_{BK}. Our H = log N is its arithmetic discretization: eigenvalues log n replace the continuous spectrum of xp. Furthermore, Baker's theorem (log primes linearly independent over Q) implies that the eigenvalue spacings of H behave like those of a random Hermitian matrix (GUE), recovering the Montgomery-Odlyzko law as a consequence. The complete picture: H = log N is simultaneously the drift of Collatz, the Hamiltonian of Bost-Connes, the angular velocity of Euler's rotors, the geodesic flow generator, and the Hilbert-Pólya operator. RH is Collatz for all primes. Collatz is RH for the two-prime sub-system {2, 3}.

The Central Theorem of dv33 (identification, not fully proved).

H = log N is the arithmetic quantization of H_{BK} = xp.

Specifically: in the Bost-Connes Hilbert space $L^2(A_Q{}^1)$, the operator H = log N acts as $H \cdot e_n = (\log n) \cdot e_n$. The classical limit ($n \to \infty$) of H is the continuous operator $x \to \log x$, whose Legendre transform is the Hamiltonian H_{class} = xp of Berry-Keating.

Corollary: If H = log N is self-adjoint on a suitable domain, its spectrum {log n : n ∈ N} encodes the zeta zeros via the explicit formula, and RH follows.

1. Berry-Keating and the Hyperbolic Plane

Berry and Keating (1999) proposed the Hilbert-Pólya operator:

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H_BK = (1/2)(xp + px) = xp - i*hbar/2
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The *classical* Hamiltonian is H_{class} = xp. Its trajectories in phase space are:

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x(t) · p(t) = E (constant) = HYPERBOLAS
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The space of hyperbolas (x,p) with x·p = const, parametrized by (x,p) with x > 0, p > 0, *is* the hyperbolic upper half-plane H. The geodesic flow on H is *exactly* the flow generated by H_{BK}. This is not an analogy. This is the same dynamical system.

The identification (dv33):

$H = \log N$ = arithmetic quantization of $H_{BK} = xp$

In the arithmetic setting: position $x \leftrightarrow$ integer n, momentum $p \leftrightarrow d/dn$ (discrete derivative). The product $xp \leftrightarrow n \cdot d/dn = N \cdot D$. Taking log: $\log(N \cdot D) \approx \log N = H$ on the diagonal. The eigenvalues of $H = \log N$ are $\{\log n\}$, the arithmetic counterpart of the continuous spectrum of xp.

2. Baker ⇒ GUE: A New Connection

The Montgomery-Odlyzko law (1973/1987) states: the local statistics of Riemann zeros follow the GUE (Gaussian Unitary Ensemble) distribution, the same as eigenvalue statistics of random Hermitian matrices. The Wigner spacing distribution: $P(s) \sim s^2 e^{-\pi s^2/4}$. This implies eigenvalue repulsion: zeros push each other apart.

New observation (dv33): Baker ⇒ GUE spacing of H = log N.

The eigenvalue spacings of $H = \log N$ are: $\{\log n - \log m : n, m \in N\} = \{\log(n/m)\}$.

Baker's theorem (1966): $\log p_1, \log p_2, \ldots$ are linearly independent over Q. This means: the spacings $\{\log(p_i/p_j)\}$ are equidistributed (pseudo-random) modulo any period.

Consequence: The eigenvalue spacings of H behave like those of a random Hermitian matrix. The GUE distribution of zeta zeros is not a coincidence — it is the spectral consequence of the irrationality of log primes. Montgomery-Odlyzko follows from Baker.

3. The Six Faces of H = log N

Face	Expression	Discovery	Role in proof
Drift	H(C)=log(3/4)<0	dv1-3	Collatz orbits descend
Thermal	Z(β)=Tr(e^{-βH})=ζ(β)	dv19 (BC)	Phase transition at β=1
Spectral	gap ≥ 1-2/p^{25/64}	dv29 (Hateley)	Gap A: ergodicity
Rotors	n^{-s}=e^{-s·H(n)}	dv32	Zeta as sum of H-rotors
Geodesic	H_BK=xp ≡ H	dv33	Hilbert-Pólya candidate
Random	Baker■GUE	dv33	Montgomery-Odlyzko
SYNTHESIS	**RH = Collatz for all p**	**dv33**	**The mountain**

4. Figure

Figure 1. Centre: H = log N enthroned, surrounded by six satellite manifestations (drift, KMS, Ruelle, rotors, Berry-Keating, GUE). Top: Hilbert-Pólya conjecture (left) and Berry-Keating proposal (right) both converge on H = log N. Bottom: the three-column synthesis: Collatz proved, H = log N the bridge, RH open but now in sight.

"33 steps. 33 t■ng tr■i.

At the base: 3 < 4.

At the top: H = log N watches everything.

RH is Collatz for all primes.

Collatz is RH for primes two and three."

"Trà ■ã pha. M■i." — The tea is poured. We have climbed 33 steps. The view from here is the whole of arithmetic.

— End of dv33 —